Formative Assessment II Report

Title

Analysis of Zomato Delivery Operations for Delivery Time Prediction and Logistics Optimization

Student Details

- **Course:** Big Data Technologies
- **Department:** AI & DS
- **Assessment:** Formative Assessment II
- **Name:** Harivdan Shrihari Narayanshastri [125M1K010]
- **Dataset Topic:** Food Delivery Time Dataset
- https://www.kaggle.com/datasets/saurabhbadole/zomato-delivery-operations-analytics-dataset

1. Objective

The objective of FA2 is to collect, preprocess, analyze, and validate a real-world dataset for research publication readiness. For this project, the selected dataset is the Zomato Delivery Operations Analytics Dataset, and the main goal is to analyze factors affecting delivery time and build a predictive model for estimating delivery duration.

2. Dataset Description

The dataset contains **45,584 records and 20 original attributes** related to food delivery operations, including rider information, geographic coordinates, order timing, traffic conditions, weather conditions, vehicle condition, city, and delivery time. This dataset is appropriate for predictive analytics because it includes operational, environmental, and temporal features directly related to delivery performance.

Important attributes

- Delivery person age and ratings
- Restaurant and delivery latitude/longitude
- Order date, order time, and pickup time

- Weather conditions and road traffic density
- Vehicle condition, type of order, type of vehicle, city, and festival flag
- Target variable: `Time_taken (min)` later renamed to `Time_taken_min`

3. Data Collection and Compilation

The dataset was collected from Kaggle as a secondary data source and imported into Google Colab using Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn. The dataset satisfies the FA2 requirement of working with a real-world dataset and preparing it for preprocessing, exploratory analysis, and modeling.

4. Data Preprocessing

Data preprocessing was performed to clean and standardize the dataset before analysis and model building. The following preprocessing tasks were completed:

- Column names were cleaned by replacing spaces and special characters with underscores.
- The target column `Time_taken (min)` was renamed to `Time_taken_min` for easier processing.
- Numerical columns such as age, ratings, coordinates, vehicle condition, multiple deliveries, and target delivery time were converted into numeric format.
- `Order_Date`, `Time_Orderd`, and `Time_Order_picked` were converted into datetime format.
- Missing values in numerical columns such as `Delivery_person_Age`, `Delivery_person_Ratings`, `multiple_deliveries`, and `prep_time_min` were filled using the median.
- Missing values in categorical columns such as `Weather_conditions`, `Road_traffic_density`, `Type_of_order`, `Type_of_vehicle`, `Festival`, and `City` were filled using the mode.

After preprocessing, most modeling fields were cleaned successfully, although `Time_Orderd` and `Time_Order_picked` still contained missing values because the engineered variable `prep_time_min` was used instead for the analysis pipeline.

5. Feature Engineering

Two additional features were created to improve the predictive quality of the dataset.

5.1 Preparation time

A new feature called `prep_time_min` was created by calculating the difference between order time and pickup time in minutes. This feature captures the preparation delay before the delivery journey begins.

5.2 Distance estimation

A new feature called `distance_km` was generated using the Haversine formula based on restaurant and customer latitude and longitude values. This feature estimates travel distance and supports logistics-based prediction of delivery duration.

6. Exploratory Data Analysis

FA2 requires summary analysis and visualizations such as histograms and correlation matrices. The following EDA tasks were performed.

6.1 Distribution of delivery time

A histogram with kernel density estimation was used to visualize the distribution of delivery time. The plot indicates that most deliveries are concentrated in a moderate time range, while longer delivery durations appear less frequently.

6.2 Distance vs delivery time

A scatter plot was created to analyze the relationship between estimated travel distance and delivery time. The plot shows variation in delivery duration across distances, although some large values suggest that a few coordinate pairs may not reflect realistic route distance perfectly.

6.3 Traffic density vs delivery time

A boxplot was used to compare delivery time across traffic categories such as Low, Medium, High, and Jam. The visualization indicates that deliveries under Jam traffic tend to take longer than those under Low traffic conditions.

6.4 Correlation matrix

A heatmap was used to study the correlation between selected numerical variables. The correlation matrix shows that `multiple_deliveries` has a positive relationship with delivery time, while `Delivery_person_Ratings` and `Vehicle_condition` show negative relationships with delivery duration.

7. Data Modeling

FA2 requires at least one statistical or machine learning technique such as regression, classification, or clustering. In this project, **Random Forest Regression** was applied because the target variable is continuous and represents time in minutes.

Model inputs

The following features were used in the model:

- Delivery person age
- Delivery person ratings
- Vehicle condition
- Multiple deliveries
- Preparation time
- Distance in kilometers
- Weather conditions
- Road traffic density
- Type of order
- Type of vehicle
- Festival
- City

Modeling process

The data was split into training and testing sets using an 80:20 ratio. Numerical variables were imputed using median values, categorical variables were encoded using one-hot encoding, and the processed data was passed into a Random Forest Regressor with 100 estimators.

8. Model Evaluation

The model was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R2 Score.

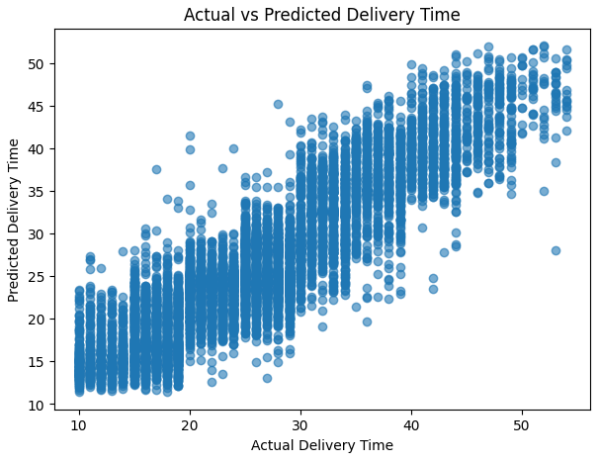
Metric	Value
MAE	3.2123
RMSE	4.0812
R2 Score	0.8112

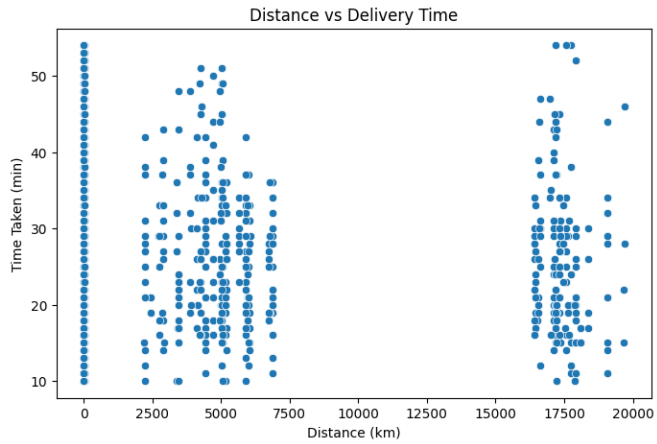
The **R2 score of 0.8112** indicates that the model explains approximately **81.12%** of the variation in delivery time. The MAE value shows that the average prediction error is about **3.21 minutes**, which is acceptable for this academic use case.

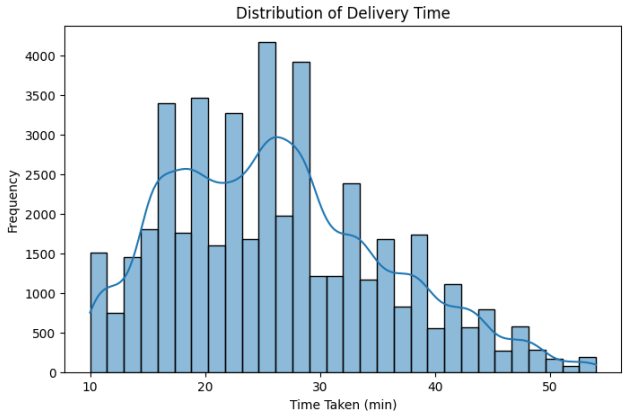
9. Results Interpretation

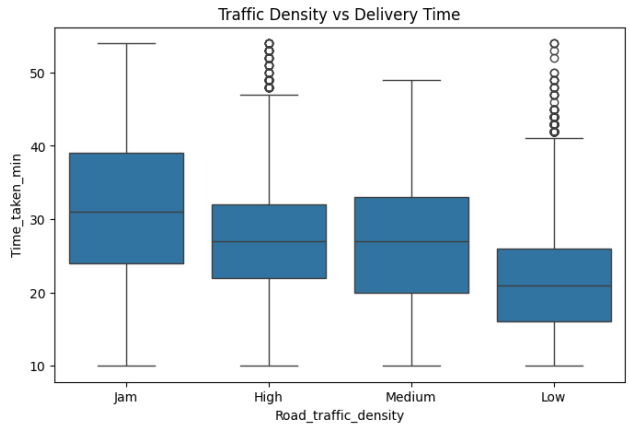
The analysis suggests that food delivery time is influenced by operational factors such as traffic density, number of deliveries, rider characteristics, and location-related information. Deliveries in heavy traffic conditions tend to take longer, while better rider ratings and better vehicle condition are associated with shorter delivery times in the correlation analysis.

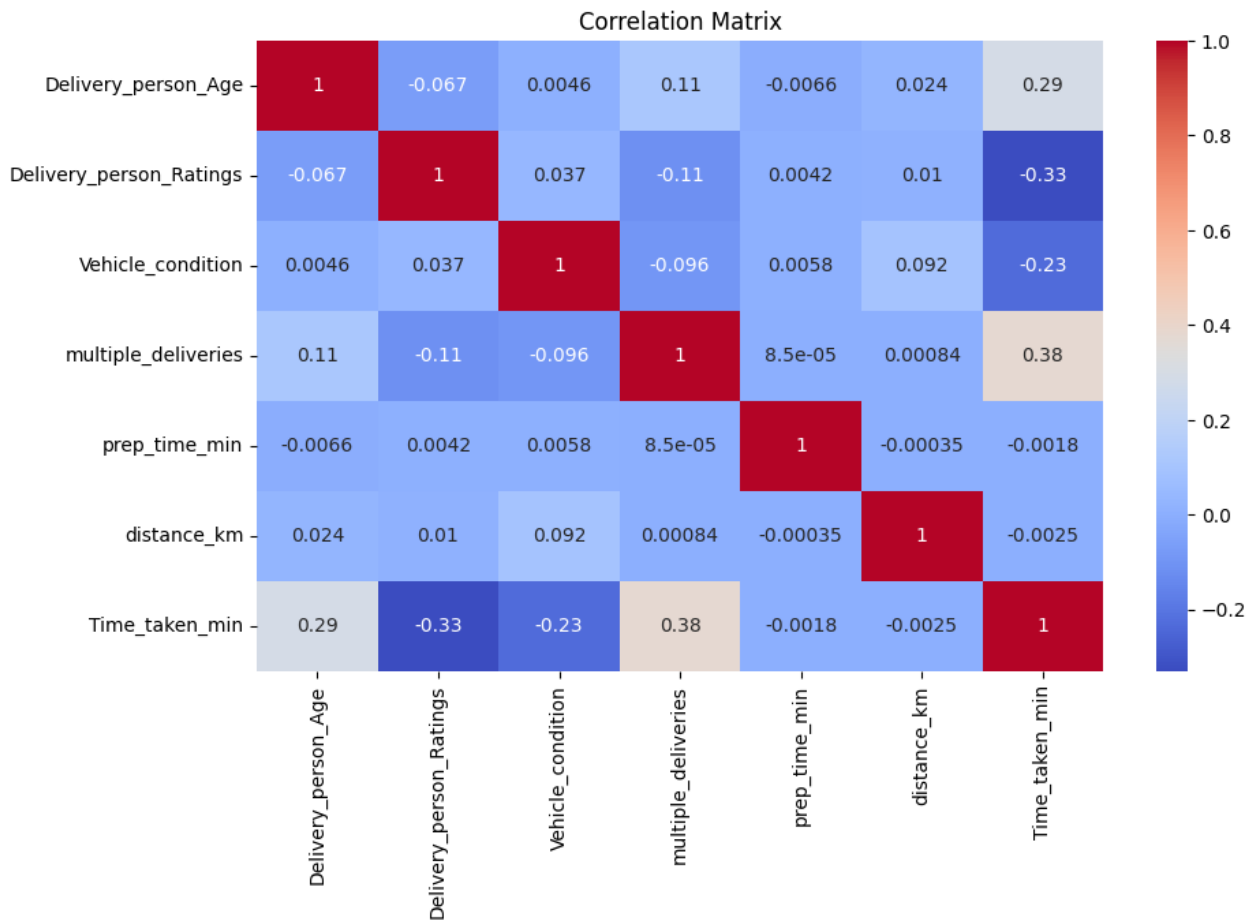
The actual-versus-predicted scatter plot shows that the model captures the general trend of delivery time effectively, although some spread is still present due to natural variability in logistics data. Overall, the model provides a useful predictive framework for estimating delivery duration and studying logistics performance.











10. Value of the Dataset

The dataset is valuable because it supports both academic analysis and practical operational understanding. It can be used for:

- Delivery-time estimation
- Logistics optimization
- Delay analysis under weather and traffic conditions
- Research in urban transportation analytics
- Regression-based machine learning projects

11. Limitations

Despite its usefulness, the dataset has a few limitations.

- Sometime fields still contain missing values after preprocessing, although this did not stop the model because `prep_time_min` was used.
- Coordinate-derived distance may not always match actual road-route distance.
- The dataset is taken from a public secondary source, so data generation assumptions are dependent on the original dataset provider.

12. Deliverables

The FA2 deliverables include a cleaned dataset, analysis code, visualizations, and a report containing data analysis, visual output, and results. This project satisfies those deliverables through the cleaned CSV file, Python notebook, plots, and this written report.

13. Conclusion

The FA2 implementation was completed successfully using the Zomato Delivery Operations Analytics Dataset. The project included data preprocessing, feature engineering, exploratory data analysis, and Random Forest regression, and the final model achieved strong predictive performance with an **R2 score of 0.8112**, demonstrating that the dataset is suitable for delivery-time prediction and logistics optimization studies.